Firas **Ben Kraiem**

Software Engineer
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Personal Profile

Software Engineer graduated from ESPRIT (2026), versatile and passionate about software development, DevOps, and artificial intelligence. I have solid experience in application development, distributed architectures, automation, and AI solution integration. Rigorous, curious, and adaptable, I am eager to leverage my technical skills within a team and contribute to building innovative and robust solutions.

Education

Private Higher School of Engineering and Technology - ESPRIT
Tunis, Tunisia

Engineering Degree in Computer Science (Specialty: Software Engineering)
Sep 2023 - Oct 2026

Higher Institute of Computer Science of Mahdia - ISIMA
Mahdia, Tunisia

Bachelor's Degree in Software Engineering and Information Systems
Sep 2020 - Jun 2023

Ibn Sina High School
Médenine, Tunisia

Baccalaureate in Computer Science
Jun 2020

Professional Experience

Elyadata
Tunis, Tunisia

Final-Year Engineering Internship (Graduation Project)
February 2026 - July 2026

- ASR Benchmarking and Model Selection: evaluated NVIDIA NeMo and K2 against Whisper and Cohere across Arabic, French, and English datasets using WER, CER, and Real-Time Factor (RTF), then selected NVIDIA NeMo FastConformer based on accuracy, streaming efficiency, stability, and tooling maturity.
- Research and Experimentation: fine-tuned a natively streaming FastConformer model on TEDxTN and TuniFra, reducing WER from 98.14% (zero-shot) to 40.31% and CER from 78.33% to 23.51% on the TEDxTN test set, establishing the final baseline for real-time Tunisian Arabic ASR.
- Real-Time Application: developed ElyaSpeech, a microservices-based transcription platform using Angular, FastAPI, and NeMo ASR. Implemented an initial WebSocket-based streaming architecture, then migrated to WebRTC for browser-to-backend audio streaming and gRPC/Protobuf for backend-to-ASR communication to improve real-time performance.
- Storage, Architecture and Deployment: integrated PostgreSQL for transcription data and Azurite (Azure Blob Storage) for audio files and model artifacts, applied hexagonal architecture, and containerized the application using Docker and Docker Compose.
- Testing, Observability and CI/CD: implemented automated unit, end-to-end, and performance testing using Pytest, Vitest, Playwright, and Locust, alongside Prometheus, Grafana, and Loki for observability, and automated quality checks and image publishing using GitHub Actions and Docker Hub.
- **Technologies**: Python, PyTorch, NVIDIA NeMo, K2, FastAPI, WebSocket, WebRTC, gRPC/Protobuf, Angular, PostgreSQL, Azurite, Docker, Docker Compose, GitHub Actions, Prometheus, Grafana, Loki, Git.

Linagora Tunisia
Tunis, Tunisia

Engineering Internship
July 2025 - August 2025

- Implemented multi-tenancy in the APISIX Dashboard: instance management, ETCD prefix-based isolation, and instance-level access control.
- Frontend in React with an instance selector to switch between multiple APISIX instances, and backend in Go with ETCD prefix injection via middleware and per-instance caching.
- Added instance monitoring (status, key metrics, Prometheus integration), JWT-based security, and Docker/Docker Compose containerization.
- **Technologies**: React, Go, APISIX, ETCD, JWT, Prometheus, Docker, Docker Compose, Git.

Devaxon
Médenine, Tunisia

Corporate Immersion Internship
July 2024 - August 2024

- Developed an equipment management application: calibration tracking, service providers, and reports, with automatic alerts for calibration dates.
- Containerized the project with Docker to simplify deployment.
- **Technologies**: Angular, TypeScript, Spring Boot, PostgreSQL, Docker, Git.

Elyadata
Tunis, Tunisia

Final-Year Internship (Bachelor's Degree)
February 2023 - June 2023

- Developed a mobile application that digitizes and automatically extracts information from ID cards, business cards, and invoices, including Arabic documents, using a multilingual OCR engine.
- Security: temporary local encryption and automatic deletion after processing, and a guided workflow (capture, cropping, preview, export).
- **Technologies**: Flutter, Dart, OCR, Swagger, Git.

Phylia Technology
Médenine, Tunisia

Summer Internship
August 2022

- Developed a meal-ordering mobile application with real-time order tracking, Firebase (authentication, real-time database, push notifications), and advanced filtering.
- **Technologies**: Flutter, Dart, Firebase, Git.

DevUp

Médenine, Tunisia

Training Project — *February 2020*

- Developed an Android weather forecast application with a modern XML interface following Android standards.
- **Technologies**: Java, XML, Android Studio.

Academic Projects

Food Finder

Tunis, Tunisia

Mobile application for discovering and booking culinary establishments — *October 2025 - November 2025*

- Mobile app to discover, book, rate, and manage favorite restaurants, with multi-role authentication (Google Sign-In, biometrics) and full CRUD on restaurants, dishes, bookings, and reviews.
- Integrated an AI assistant (Gemini) for contextual recommendations. Modular architecture (presentation, services, SQLite persistence, FastAPI backend) and a reusable UI library (filters, loaders, cards, dialogs).
- **Technologies**: Flutter, Dart, FastAPI, Python, SQLite, AI, Git.

AI Chatbot Assistant

Tunis, Tunisia

Full-Featured Conversational AI Application — *October 2025*

- Full-featured conversational AI application with complete conversation history management and CRUD operations, integrating the OpenAI GPT API for real-time chat and message regeneration.
- Scalable RESTful backend with Spring Boot and a modern, responsive UI built with Angular 19, PrimeNG, and Tailwind CSS.
- **Technologies**: Angular, Spring Boot, TypeScript, PrimeNG, Tailwind CSS, REST API, OpenAI API.

DevOps

Tunis, Tunisia

Hands-on practice with key DevOps tools — *September 2025 - October 2025*

- Hands-on practice with the main DevOps tools: orchestration, CI/CD, monitoring, code quality, and containerization.
- **Technologies**: Kubernetes, Jenkins, Prometheus, SonarQube, Docker.

Coding Factory

Tunis, Tunisia

Academic Management Platform — *December 2024 - May 2025*

- Platform managing final projects (PFE), courses, events, and technical forums, with AI features: course recommendations and training duration prediction.
- Microservices architecture (Eureka Server, API Gateway, Feign Client, Spring Cloud Config) with modular deployment using Docker Compose.
- **Technologies**: Angular, Spring Boot, TypeScript, Tailwind CSS, Neon PostgreSQL, Docker, Docker Compose, ML, Git.

CareLink

Tunis, Tunisia

Cross-Platform Client-Medical Application — *February 2024 - April 2024*

- Application for comprehensive patient management (appointments, medical records, follow-up), with a distributed architecture across three machines (application, database, clients) to enhance security and communication.
- **Technologies**: Symfony (Web), FlutterFlow (Mobile), JavaFX (Desktop), Git.

Technical Skills

- **Programming Languages:** C / C++, C#, Java, Python, Dart, Go, JavaScript, TypeScript, PHP, SQL, LINQ, PL/SQL
- **Web / Frameworks and Libraries:** HTML, CSS, Bootstrap, Tailwind CSS, Angular, React, Symfony / Twig, .Net / ASP .Net, Spring Boot, Flutter, FastAPI, Flask
- **Microservices, Real-Time and Gateways:** Eureka, API Gateway, Feign Client, Spring Cloud Config, WebSocket, WebRTC, gRPC / Protobuf, Apache APISIX
- **Databases:** MySQL, Oracle, SQLite, PostgreSQL, Neon PostgreSQL, Firebase, MongoDB, ETCD, phpMyAdmin
- **DevOps, Containerization and Observability:** Docker / Docker Compose, GitHub Actions, Jenkins (CI/CD), Kubernetes, MiniKube, SonarQube, Prometheus, Grafana, Loki
- **AI / Machine Learning:** Machine Learning, Deep Learning, PyTorch, NVIDIA NeMo, K2 / icefall, Speech Recognition (ASR), Model Fine-tuning, Hugging Face, scikit-learn
- **Testing, API and Documentation:** Pytest, Vitest, Playwright, Locust, Swagger, Postman
- **Version Control and Project Management:** Git, GitHub, Trello, Jira
- **Modeling, Design and Description Languages:** Merise, UML, XML, FXML
- **Systems, Scripting and Network Security:** Linux, Windows, Shell, IDS/IPS, Network Security

Software

- **IDEs and Editors:** VS Code, IntelliJ IDEA, Android Studio, NetBeans, CodeBlocks, Python IDLE
- **Modeling, Documentation, Virtualization and Cloud:** StarUML, Overleaf LaTeX, VMware Workstation, VirtualBox, Google Colab, Kaggle, Docker Desktop / Docker Hub
- **BI and No-Code / Low-Code:** Power BI, FlutterFlow, n8n

Languages

- Arabic (Native) | French (Advanced) | English (Intermediate/Good)

Associative Activities

- **IEEE ISIMA Student Branch**, Mahdia, Tunisia | Member | **Microsoft Club**, Mahdia, Tunisia | Member